

Mohamad Chehade

Research Curriculum Vitae

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Research interests: risk-aware reinforcement learning and transfer, large language model alignment, time-series foundation models, learning for optimization and control, and neural-network verification and certified robustness.

Education

The University of Texas at Austin

Ph.D., Electrical and Computer Engineering

Master of Science in Engineering, Aug. 2023 – May 2026. Advisor: Dr. Hao Zhu.

Austin, TX

Aug. 2023 – Present

American University of Beirut

B.Eng., Electrical and Computer Engineering

GPA: 4.0/4.0; Minor in Mathematics. Final-year project: Optimal Power Flow via Machine Learning (Dr. Rabih Jabr).

Undergraduate research: Microgrid Sizing using Ordinal Optimization (Dr. Sami Karaki).

Beirut, Lebanon

Aug. 2019 – Jun. 2023

Skills and Relevant Coursework

Programming: Python, C/C++, MATLAB, SQL, R

ML and scientific computing: PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Matplotlib

Optimization: Gurobi; convex and mixed-integer optimization, ordinal optimization

Power systems: PowerWorld, MATPOWER, Simulink, SPICE, HOMER, PVSyst, LabVIEW

Development: Git, Linux, LaTeX, Overleaf, Microsoft Office Suite

Languages: English (fluent), French (fluent), Arabic (native)

Coursework: Reinforcement Learning; Generative AI; Statistical Machine Learning; Applied Machine Learning; Probability and Stochastic Processes; Applied Stochastic Processes; Learning-based Optimal Control; Convex Optimization; Energy Optimization and Operation

Research and Engineering Experience

The University of Texas at Austin

Graduate Research Assistant – Advisor: Dr. Hao Zhu

Austin, TX

Aug. 2023 – Present

- **Risk adaptation (TRAM):** Developed test-time policy composition to adjust reward–risk trade-offs without re-training source agents.
- **LLM alignment (SITAlign):** Developed inference-time alignment that optimizes a primary objective subject to thresholds on other criteria.
- **LLMs for grid actions:** Fine-tuned language models with supervision and preferences to propose feasible corrective switching plans.
- **Distribution-grid control (NEO-Grid):** Built neural power-flow surrogates and deep equilibrium models for voltage optimization and control.
- **Battery control:** Compared model-free and model-based controllers for battery operation under time-varying electricity prices.
- **Carbon-aware energy management:** Co-developed market-participation methods that weigh building energy costs against carbon emissions.

Argonne National Laboratory

Research Aide – Grid Optimization and Infrastructure Group

Mentors: Dr. Feng Qiu and Dr. Wei Gao.

Remote

May – Aug. 2026

- Developed decision-aware time-series foundation models to align forecasts with downstream decision-making and optimization.

Los Alamos National Laboratory

Graduate Research Assistant – T-5 Applied Mathematics and Plasma Physics

Mentor: Dr. Russell Bent.

Remote

May – Aug. 2025

- Developed robustness-training methods that use verifiable input regions to improve neural-network certified robustness.

Los Alamos National Laboratory

Graduate Research Assistant – T-5 Applied Mathematics and Plasma Physics

Mentors: Dr. Wenting Li and Dr. Brian Bell.

Los Alamos, NM

Jun. – Aug. 2024

- Developed two algorithms for large exact verifiable input regions of neural networks, targeting physical and safety-critical systems (LEVIS, ICML 2025).

- Developed inverter fault diagnosis using combinational logic with industrial motor manufacturer Ward Leonard.
- Optimized and built an inverter circuit for high-power testing using mixed-integer linear programming.

Publications – Published or Accepted

1. **M. Chehade**, A. S. Bedi, S. Chakraborty, A. Zhang, and H. Zhu. Test-Time Risk Adaptation with Mixture of Agents. *NeurIPS 2026 (accepted)*.
2. **M. Chehade**, W. Li, B. W. Bell, R. Bent, S. R. Kazi, and H. Zhu. LEVIS: Large Exact Verifiable Input Spaces for Neural Networks. *ICML 2025*.
3. **M. Chehade**, S. S. Ghosal, S. Chakraborty, A. Reddy, D. Manocha, H. Zhu, and A. S. Bedi. Bounded Rationality for LLMs: Satisficing Alignment at Inference-Time. *ICML 2025*.
4. **M. Chehade** and H. Zhu. Fine-Tuning LLMs to Generate Economical and Reliable Actions for the Power Grid. *IEEE PES General Meeting 2026 (accepted)*.
5. **M. Chehade** and H. Zhu. NEO-Grid: A Neural Approximation Framework for Optimization and Control in Distribution Grids. *Hawaii International Conference on System Sciences (HICSS), 2026*.
6. Y.-H. Cho, **M. Chehade**, F. Al-Janahi, S. Lim, J. Mohammadi, and H. Zhu. Carbon-aware Market Participation for Building Energy Management Systems. *IEEE PES General Meeting 2026 (accepted)*.
7. **M. Chehade** and S. Karaki. BOOST: Microgrid Sizing Using Ordinal Optimization. *IEEE Texas Power and Energy Conference (TPEC), 2025, pp. 1–4*.
8. **M. Chehade**, Y.-H. Cho, S. Chinchali, and H. Zhu. Should We Use Model-Free or Model-Based Control? A Case Study of Battery Control. *North American Power Symposium (NAPS), 2024, pp. 1–5*.

Awards and Honors

2026 Third place, American Society of Mechanical Engineers (ASME) CIE Student Hackathon – AI for engineering design and manufacturing.
2024 Best Graduate Presentation Award, North American Power Symposium (NAPS).
2023 Mohamad Ali Safieddine Award for Academic Excellence – ranked first across the American University of Beirut Faculty of Engineering and Architecture.
2023–present Cockrell School of Engineering Fellowship, UT Austin.
2023 Electrical and Computer Engineering Distinguished Graduate Award – ranked first among ECE graduates.
2023 Exceptional ECE Final Year Project Award – Power and Energy Systems.
2021 Best Pitch Award, EPFL Tech4Impact Summer School (Swiss Federal Institute of Technology Lausanne).
2019–2023 Dean's Honor List, American University of Beirut – every awarded semester.

Tutorial

Upcoming, October 26, 2026 – IEEE SmartGridComm, College Station, TX. Agentic AI for Grid Operators in Active Distribution Networks within a Multi-Agent Framework. Co-presenters: Alaa Selim (Dartmouth College) and Mahmoud Abdelaal (ABB).

Peer Review Service

Neural Information Processing Systems (NeurIPS)	June 2026 – Present
International Conference on Machine Learning (ICML)	Feb. 2026 – Present
International Conference on Learning Representations (ICLR)	Oct. 2025 – Present
AAAI Conference on Artificial Intelligence	Aug. 2026 – Present
IEEE Transactions on Smart Grid	Sep. 2023 – Present
IEEE Power Engineering Letters	May 2026 – Present
IEEE Systems Journal	Sep. 2025 – Present
Hawaii International Conference on System Sciences (HICSS)	June 2025 – Present
Asilomar Conference on Signals, Systems, and Computers	July 2024 – Present
IEEE Texas Power and Energy Conference (TPEC)	Dec. 2024 – Present